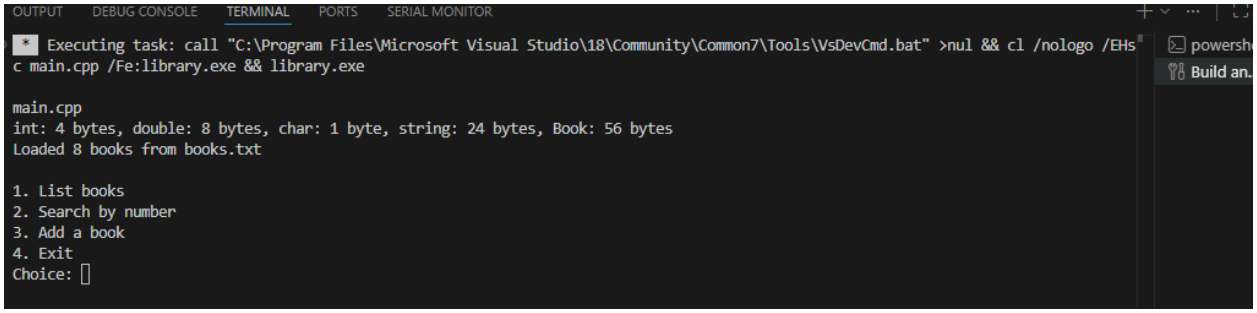
Program Explanation

This program is a small library catalog of books that have sold more than 100 million copies. When it starts, it prints the sizes of the main data types and loads the books from a text file called books.txt. A menu then lets the user list all the books, search for a book by its number, add a new book, or exit. The program has two classes. The Book class stores one book's number, title, author, and copies sold. The Library class holds an array of up to 20 books. It reads books.txt, where each line is one book in the format number|title|author|sales. It prints the list, adds new books, and searches for a book with a binary search that returns a pointer to it. Every new book gets the next number, so the array stays in order for the binary search. It is also written to books.txt right away, so it is still there the next time the program runs.

Concepts Used

- Data types and data sizes: int, double, char, string, and sizeof
- File types: books.txt
- File I/O: ifstream to read and ofstream to write
- Pointers: Book* returned by the search function
- Arrays: Book books[20]
- Binary search: search by book number
- Strings: title and author, getline, and find
- Two classes: Book and Library

Main menu



List books

```
main.cpp
int: 4 bytes, double: 8 bytes, char: 1 byte, string: 24 bytes, Book: 56 bytes
Loaded 8 books from books.txt

1. List books
2. Search by number
3. Add a book
4. Exit
Choice: 1
1. A Tale of Two Cities - Charles Dickens - 200 million copies sold
2. Alice's Adventures in Wonderland - Lewis Carroll - 100 million copies sold
3. And Then There Were None - Agatha Christie - 100 million copies sold
4. Dream of the Red Chamber - Cao Xueqin - 100 million copies sold
5. Harry Potter and the Philosopher's Stone - J. K. Rowling - 120 million copies sold
6. The Alchemist - Paulo Coelho - 150 million copies sold
7. The Hobbit - J. R. R. Tolkien - 100 million copies sold
8. The Little Prince - Antoine de Saint-Exupery - 200 million copies sold
```

Add a book

```

1. List books
2. Search by number
3. Add a book
4. Exit
Choice: 3
Title: The Catcher in the Rye
Author: J. D. Salinger
Copies sold (millions): 65
Added as book 9 and saved to books.txt

1. List books
2. Search by number
3. Add a book
4. Exit
Choice: 1
1. A Tale of Two Cities - Charles Dickens - 200 million copies sold
2. Alice's Adventures in Wonderland - Lewis Carroll - 100 million copies sold
3. And Then There Were None - Agatha Christie - 100 million copies sold
4. Dream of the Red Chamber - Cao Xueqin - 100 million copies sold
5. Harry Potter and the Philosopher's Stone - J. K. Rowling - 120 million copies sold
6. The Alchemist - Paulo Coelho - 150 million copies sold
7. The Hobbit - J. R. R. Tolkien - 100 million copies sold
8. The Little Prince - Antoine de Saint-Exupery - 200 million copies sold
9. The Catcher in the Rye - J. D. Salinger - 65 million copies sold
```